

[USB-IF Compliance Updates](#)

Number of Updates: 23

Table of Contents					
ID	Updated	Subject	Reason	Mandate	Effective Date
165	May, 2025	Full-Speed and Low-Speed Devices - 5 Tier Hub Signal Quality Test Deprecation	Electrical Signal Quality testing behind 5 tiers of hubs is no longer mandatory for Full-Speed and Low-Speed Devices	Optional	May, 2025
142	November, 2021	USB 2.0 Edge Monotonicity	Clarification of the USB 2.0 Monotonicity Requirement	Required	November, 2021
45	June, 2021	Inrush Current Test Description	Edits to this update highlighted in yellow.	Required	June, 2021
86	March, 2021	Hi-Speed Signal Quality Test Fixture	Standardization on a single electrical test fixture	Required	August, 2013
134	January, 2020	Receiver Sensitivity (EL_17) and Squelch (EL_16) Histogram measurement areas defined	Undefined measurement window for the histogram method	Required	February, 2020
3	January, 2020	Receiver Sensitivity (EL_17) and Squelch (EL_16)	Clarify squelch and receiver sensitivity parameters	Required	February, 2005
133	September, 2019	LFPS Tx Low Power/LFPS Rx Detect Threshold	Test Clarification	Required	September, 2019
59	January, 2019	(DEPRECATED) ECN: Minimum Capacitance Required on Vbus	(DEPRECATED) Enables Attach Detection without VBus present	Informational	December, 2008
7	March, 2015	Rise and fall time measurements for USB compliance	Change in measurement	Required	August, 2007
98	October, 2014	ECN: Increase to the Maximum Voltage on VBUS	Align the USB 2.0 Specification with the USB 3.1 and Type C Specifications	Required	August, 2014
93	May, 2014	Low-speed and Full-speed Eye Template Definition	Publication of the low- and full-speed eye templates used for certification	Informational	
92	April, 2014	Maximum allowed voltage on non-driven lines of Test_J and Test_K are relaxed	Relaxation of compliance requirements for hi-speed EL_9 test	Required	April, 2014
41	August, 2013	USBET20, version 1.20 published	Announcement of update to the USB Electrical Analysis Tool, USBET	Required	April, 2013
87	March, 2013	New Rise and Fall Time Parameters for High-speed Signaling	Relaxation of the hi-speed rise and fall time parameters	Required	April, 2013
81	August, 2011	64-bit HSET is Unable to Initialize and Execute Tests.	"No Host Controllers Found" error message	Informational	September, 2010
67	February, 2010	The Driven Lines of High-speed Test_J and Test_K are Removed as Certification Requirements	EL_8, Driven Line Test_J and Test_K are obsolete	Optional	January, 2010
61	March, 2009	ECN: Host Controllers must Assure SOFs are in Phase -Lock	Enables isochronous peripherals to take advantage of aggressive power management	Required	December, 2008
4	May, 2008	Engineering Change Notice: Maximum Suspend Current	Relaxation of suspend current requirements for USB peripherals	Required	April, 2008
43	October, 2007	EL_22 - Interpacket Delay for Hosts with Embedded Hubs	Defines Interpacket Timing for hosts with embedded hubs	Informational	n/a

ID	Updated	Subject	Reason	Mandate	Effective Date
34	July, 2007	Peripherals may support both bus-power and self-power modes	Clarification on how to declare and use both bus-power and self-power modes	Informational	n/a
8	June, 2007	"Near-end" and "Far-end" explained	Clarification	Informational	
5	February, 2007	High-speed Devices: Average current is to be measured at both HS and FS	Test Procedure Clarification	Required	Now
6	June, 2006	Disconnect Test no longer a compliance requirement for downstream ports	Test Change to the Compliance Program	Informational	June, 2006

LFPS Tx Low Power/LFPS Rx Detect Threshold

Mandate: Required

Effective Date: September, 2019

USB3.2 Transmitter specification Rev1.0 (section 6.7.2) defines a Low Power Transmitter operation.

In addition to the full swing transmitter specification, an optional low power swing transmitter is also specified for SuperSpeed applications. A low power swing transmitter is typically used in systems that are sensitive to power and noise interference, and have a relatively short channel. The requirement as to whether a transmitter needs to support full swing, low power swing, or both swings, is dependent on its usage model. All SuperSpeed transmitters must support full swing, while support for low power swing is optional. The method by which the output swing is selected is not defined in the specification, and is implementation specific.

LFPS Tx test procedures specific for Captive Cable devices are included in the below document.

More details may be found in the document located [here](#).



[Top](#)

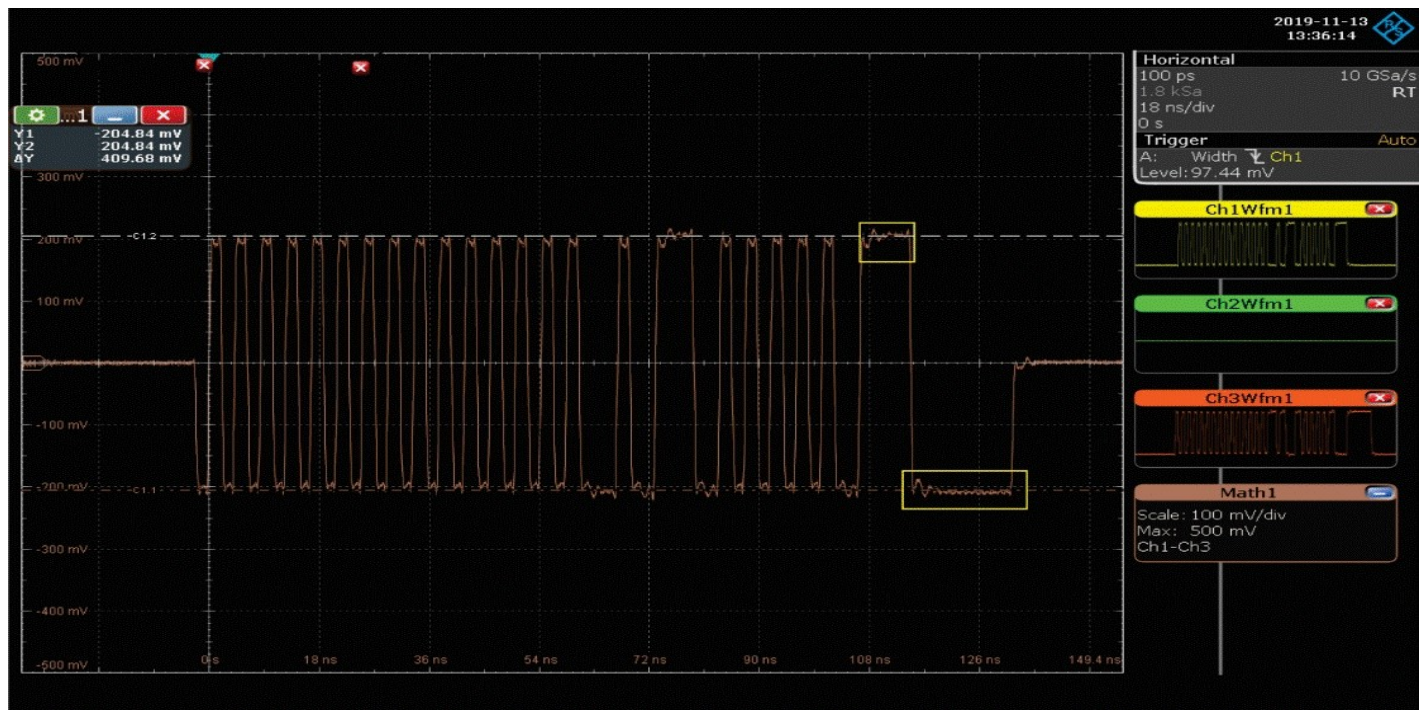
Receiver Sensitivity (EL_17) and Squelch (EL_16) Histogram measurement areas defined

Mandate: Required

Effective Date: February, 2020

Receiver Sensitivity (EL_17) and Squelch (EL_16) Histogram measurement areas defined

The histogram measurements for EL_16 and EL_17 are to be taken in the multi-bit areas at the end of the data packet as these areas are the least likely to be affected by reflections.



[↑ Top](#)

USB 2.0 Edge Monotonicity

Mandate: Required

Effective Date: November, 2021

The Edge Monotonicity requirement, shown in reports generated by the USB 2.0 electrical compliance tool USBET, was introduced in the early 2000's and derives from the hysteresis specifications of silicon's available at the time. This update clarifies these requirements.

Edge Monotonicity for Low and Full-speed products must be less than 0.5V.

Edge Monotonicity for High-speed products must be less than 50e-3V.

[↑ Top](#)

Full-Speed and Low-Speed Devices - 5 Tier Hub Signal Quality Test Deprecation

Mandate: Optional

Effective Date: May, 2025

Basic signal quality testing for Full Speed (FS) and Low Speed (LS) devices no longer needs to be performed behind five tiers of hubs.

Low Speed:

Low-speed devices must have a captive cable, so no specific cable requirement applies?they should be tested as-is.

Full Speed Cable Requirements:

- Standard-B: 5 meters
- Mini-B: 4.5 meters
- Micro-B: 2 meters
- USB-C: 4 meters

[↑ Top](#)

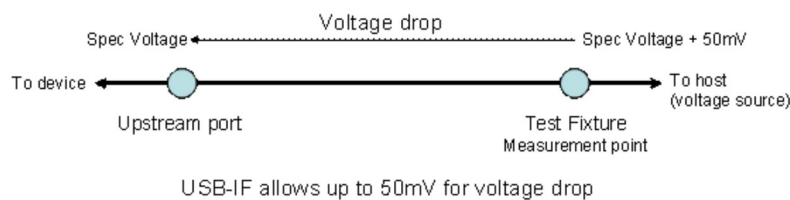
Receiver Sensitivity (EL_17) and Squelch (EL_16)

Mandate: Required

Effective Date: February, 2005

There is confusion in how to apply the automatic waiver relating to EL_16 and EL_17. As a result, devices are being incorrectly designed. Hopefully, the following will clarify the rules for EL_16 and EL_17.

The waiver is derived like this:

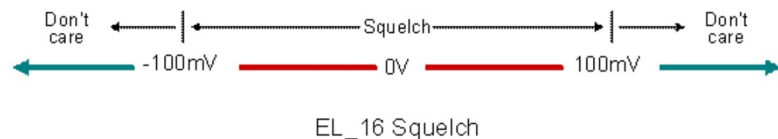


Receiver Sensitivity and Squelch measurements are supposed to be made at the upstream port pins as defined in Table 7-3 and Figure 7-12 of the USB 2.0 Specification. However, the test fixture used at USB-IF workshops and test facilities does not permit measurement at the upstream port pins. Because the voltage at the measurement point is higher than at the upstream port, the USB-IF provides a 50mV allowance for the voltage drop between the two points. Again, the voltage at the measurement point is higher than at the upstream port pins, so the 50mV allowance has to be added to the spec requirement -- not subtracted.

Voltage is differential, so there is a positive and negative voltage level. The text of the waiver was meant to add 50 to the positive voltage and add -50 to the negative voltage.

EL_16

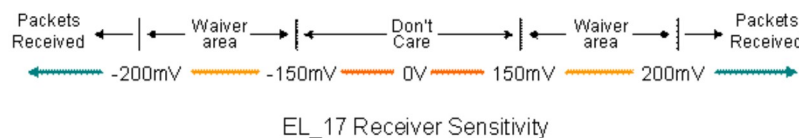
The USB 2.0 specification in Section 7.1.7.2 requires squelch (EL_16) to occur below 100mV magnitude. So no packets must be acknowledged between -100mV and +100mV. This has always been the requirement. Full squelch may occur at higher voltages, but it is mandatory between -100mV and +100mV.



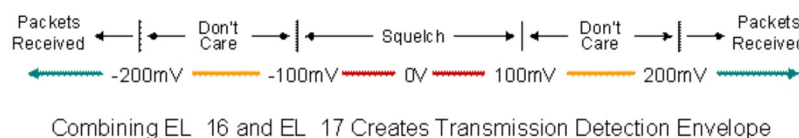
The USB-IF applies the specification requirement (100mV) to measurements made at the test fixture instead of the upstream port. So the voltage drop acts like a built-in waiver for test procedure EL_16. Thus, THERE IS NO 50mV WAIVER FOR EL_16. The HS Test Procedure is incorrect for stating a waiver for EL_16.

EL_17

Receiver sensitivity requires all packets to be reliably received down to 150mV magnitude. Packets may be received at lower voltages, but it is mandatory at levels above 150mV magnitude. Again, this measurement is to be made at the upstream pins but the test fixture does not allow this. So the USB-IF grants a generous 50mV waiver to compensate for the voltage drop away from the upstream pins. Thus, the USB-IF requires packets to be reliably received at levels above 200mV for EL_17. Packets can, but do not need to be, received between -200mV and +200mV.



So looking at EL_16 and EL_17 together with the waiver, the graph looks like this:



Combining EL_16 and EL_17 Creates Transmission Detection Envelope

Please note that it is the combination of EL_16 and EL_17 that validates the "transmission detection envelope" defined in section 7.1.4.2 of the USB 2.0 Specification.

Examples

Example 1:

A device exhibits full squelch below 300mV magnitude.

- EL_16 is a pass because full squelch occurs below 100mV magnitude.
- EL_17 fails because packets are not reliably received above 200mV with the waiver

Example 2:

A device exhibits full squelch below 110mV magnitude and receives packets above 120mV magnitude

- EL_16 is a pass because squelch occurs below 100mV magnitude
- EL_17 is a pass because packets are reliably received above 150mV magnitude

Example 3:

A device exhibits full squelch below 90mV and receives packets above 120mV magnitude

- EL_16 is a fail because packets are received below 100mV magnitude
- EL_17 is a pass because packets are reliably received above 150mV magnitude

Example 4:

A device exhibits full squelch below 160mV and receives packets above 190mV magnitude

- EL_16 is a pass because squelch occurs below 100mV magnitude
- EL_17 is a waiver because packets are reliably received within 50mV of 150mV magnitude

Example 5:

A device exhibits full squelch below 180mV and receives packets above 210mV magnitude

- EL_16 is a pass because squelch occurs below 100mV magnitude
- EL_17 is a fail because packets are not reliably received within 50mV of 150mV magnitude



[Top](#)

Peripherals may support both bus-power and self-power modes

Mandate: Informational

Effective Date: n/a

A peripheral that draws less than 100mA from USB may report itself as either bus-powered or self-powered in the `bmAttributes` field of the active configuration descriptor. Regardless of the power source, the average current drawn from the bus must never exceed the value declared in the `bMaxPower` field of the active configuration descriptor. Also, the `GetStatus(DEVICE)` call must accurately reflect whether the peripheral is operating under bus-power or self-power.

A device that is actively drawing more than 100mA from USB must report itself as bus-powered in the `GetStatus(DEVICE)` call. Peripherals that return "Self-powered" in the `GetStatus(DEVICE)` call are prohibited from drawing more than 100mA at any time.

A peripheral is allowed to support both bus and self-powered modes in a configuration. To support both bus and self-powered operation, the `bMaxPower` field is to report the worse case current demand when operating under bus-power and the self-powered bit of the `bmAttributes` is marked for self-power. In all cases, the `GetStatus(DEVICE)` call must accurately report whether the device is currently operating on self- or bus-power. Please see section 9.6.3 of the USB 2.0 Specification.

A self-powered peripheral that experiences a loss of power is not allowed to switch to a high-power, bus-powered mode without forcing a re-enumeration. The USB 2.0 Specification Section 7.2.1.2 defines this requirement for self-powered hubs that also support bus-powered operation. Thus, the USB-IF enforces this requirement on standard peripherals that support both bus- and self-powered modes as well.

The reason for requiring re-enumeration is that the peripheral will go from a low-current draw to a high-current draw. The operating system must be allowed to manage the unexpected power change of the peripheral from a low-power requirement to a high-power requirement.



[Top](#)

Engineering Change Notice: Maximum Suspend Current

Mandate: Required

Effective Date: April, 2008

An Engineering Change Notice to the USB 2.0 Specification named "Suspend Current Limit Changes" has been issued in April 2008. This ECN changes the maximum `ICCSL` value to 2.5mA.

The purpose of this ECN is to allow any USB peripheral to consume up to 2.5mA during suspend. It does not matter whether the peripheral is low-power or high-power, and it does not matter whether remote wakeup is enabled.

Please note, however, that the InterChip suspend current requirements does not change and remains at 150uA.

The "Suspend Current Limit Changes" ECN is included with the download of the USB 2.0 Specification.



[Top](#)

USBET20, version 1.20 published

Mandate: Required

Effective Date: April, 2013

The USB Electrical Analysis Tool, USBET, has a major update and is available for download. With a new name and version, USBET20, version 1.20, is the *only* official electrical analysis tool of the USB-IF. There are no changes in measurements or pass/fail criteria, however. Please see the [USB Tools webpage](#) for download information.

Once installed, documentation can be found at: **Start->All Programs->USB-IF Test Suite->USBET20->Documentation**

For best results, please use the new [Hi-Speed Signal Quality Test Fixture](#).

[↑ Top](#)

EL_22 - Interpacket Delay for Hosts with Embedded Hubs

Mandate: Informational

Effective Date: n/a

EL_22 - Interpacket Delay for Hosts with Embedded Hubs

Hosts are permitted to embed a high-speed hub and expose the hub's downstream ports. Because the embedded hub must relay the data of the host, the interpacket gap timing is affected when transmitting after receiving a packet.

Section 7.1.19.2 of the USB 2.0 Specification identifies high-speed delay times for cables and hubs. With a max propagation of 26ns for a cable and a max hub repeater delay of 36 bits +4ns (Test Spec EL_48), there is sufficient information for determining what the maximum response time to a device packet behind one tier of hubs should be.

Max delay from downstream port of hub to the downstream port of the host is:

$$\begin{array}{r} 26\text{ns} \\ 4\text{ns} + 36 \text{ bits} \\ \hline 30\text{ns} + 36 \text{ bits} \end{array}$$

Multiplied by two since it is a "round trip" yields:

$$60\text{ns} + 72 \text{ bits}$$

Add in the max turnaround time for the host, 192bits

$$\begin{array}{r} 60\text{ns} + 72 \text{ bits} \\ + 192 \text{ bits} \\ \hline 60\text{ns} + 264 \text{ bits} \end{array}$$

So the maximum interpacket delay of a host's response to a device is $\leq 264 \text{ bit times} + 60\text{ns}$, which equates to 610ns or 292 bit times.

[↑ Top](#)

Inrush Current Test Description

Mandate: Required

Effective Date: June, 2021

Inrush Current Analysis

The USB-IF requires an Inrush Current Test for all upstream USB ports. The purpose of the test is to ensure that the current consumed due to bulk capacitance and peripheral startup does not cause the voltage to drop below valid levels. A newly attached device with excessive inrush current can cause devices that are connected on adjacent ports to disconnect due to low VBus voltage. Bus powered hubs are most susceptible to voltage brownout from large inrush events.

The USB-IF Electrical Analysis Tool, USBET, revision 2.22 implements a new algorithm for assessing inrush current. The new inrush algorithm is being enforced now. Studies of this implementation show that it is easier for devices to pass the inrush test using this new algorithm. The following describes the algorithm and requirements for inrush testing.

Measurement Method

Inrush current is measured for a minimum of 100 milliseconds after attach. Attach is defined at the moment the VBus and ground pins of the plug mate with the receptacle. Any current exceeding 100 mA during the 100 ms interval is considered part of the inrush current event. The inrush current is divided into regions. A region is an interval where the current exceeds 100 mA until the time the current falls below 100 mA for at least 100 μs . There can be multiple inrush regions during the 100 ms period. Pass/fail is determined by the region having the highest charge.

Multiple Regions

A region starts when the current rises above 100 mA and ends when the current is below 100 mA *for at least 100 μs* . There may be multiple regions during the 100 ms period. The analysis tool calculates the charge above 100 mA in each region. The charge of each region is listed in the report. Pass or fail is based on the region with the largest charge. The regions are shown on the plot by vertical yellow lines marking the start and end of each region.

Minimum Sample Rate

A minimum sample rate of 1 megasamples per second is required for proper analysis. If the sample rate is below this value, the program still performs the analysis but will record an error indicating that the sample rate is too low.

Record Length

A record length of 100 milliseconds or more after the first inrush event is required for the measurement. If the data record is less than 100ms, the result will be a failure because of insufficient measurement time.

If the oscilloscope does not have sufficient memory to capture 100ms of data, then the inrush test will need to be divided into 30ms segments. Each time the inrush test is performed, the trigger should be delayed to capture the next segment. Use USBET to analyze each inrush segment. Repeat until all segments covering at least 100ms of inrush data is examined. Each and every individual inrush segment must pass. A failure in one of the inrush segments constitutes an overall failure.

No Inrush Event Found

If there is no inrush event found, *i.e.* the current never rises above 100 mA, the result is considered a Pass. However, the inrush current test should be repeated several times to ensure an accurate measurement. The USB-IF has deprecated the requirement for peripherals to have a minimum capacitance of 1uF.

USB-C Inrush Test Conditions.

USB-C DRPs must be tested in 8 configurations:

- DRP unpowered with Rp default by tester
- DRP unpowered with Rp 1.5 by tester
- DRP unpowered with Rp 3 by tester
- DRP powered with Rp default by tester
- DRP powered with Rp 1.5A default by tester
- DRP powered with Rp 3A default by tester
- DRP unpowered in the Unattached state
- DRP powered in the Unattached state

USB-C Sinks must be tested in 8 configurations:

- Sink unpowered with Rp default by tester
- Sink unpowered with Rp 1.5 by tester
- Sink unpowered with Rp 3 by tester
- Sink powered with Rp default by tester
- Sink powered with Rp 1.5A default by tester
- Sink powered with Rp 3A default by tester
- Sink unpowered in the Unattached state
- Sink powered in the Unattached state


[Top](#)

High-speed Devices: Average current is to be measured at both HS and FS

Mandate: Required

Effective Date: Now

For high-speed devices, average current draw is to be measured at both high-speed and full-speed. Current draw at high-speed tends to be larger than at full-speed, but this is not always true.


[Top](#)

(DEPRECATED) ECN: Minimum Capacitance Required on Vbus

Mandate: Informational

Effective Date: December, 2008

Effective January 2019, the following requirement is deprecated.

Revision 2.0 of the USB On-The-Go Supplement defines an Attach Detection Protocol (ADP) for hosts and peripherals. This protocol enables peripherals and hosts to detect when a device attach occurs when VBus is turned off. In order for ADP to work from a host's perspective, peripherals are required to have a minimum of 1uF (C_{RPP}) capacitance on their VBus pin.

This ECN is part of the download of the USB 2.0 Specification and should be reviewed for specific details.

[↑ Top](#)

Disconnect Test no longer a compliance requirement for downstream ports

Mandate: Informational

Effective Date: June, 2006

The disconnect test is no longer required for certification of downstream ports. Vendors are encouraged to verify disconnect voltage thresholds on their own.

The Host Disconnect Test was performed on all downstream ports (hosts and hubs). When a HS device removes its HS terminations or is detached, the voltage on the data lines increases. An increased voltage level indicates a disconnect and the port is required to signal a disconnect.

[↑ Top](#)

ECN: Host Controllers must Assure SOFs are in Phase -Lock

Mandate: Required

Effective Date: December, 2008

Host controllers are to issue SOF tokens that are in phase-lock across selective suspend or level 1 (L1) power management events. Phased-locked SOFs are not required across global suspend conditions.

This ECN is part of the download of the USB 2.0 Specification. Please see the ECN for specific details.

[↑ Top](#)

The Driven Lines of High-speed Test_J and Test_K are Removed as Certification Requirements

Mandate: Optional

Effective Date: January, 2010

The USB-IF no longer requires EL_8: Test_J and Test_K to be performed as a condition for USB Certification. USB certification of high-speed USB devices (hosts, hubs and peripherals) no longer requires execution or passing of EL_8: Test_J and Test_K tests. EL_8 is defined in the USB 2.0 Electrical Test Specification and measures the data line voltage when driven high. The nature of this test measures the DC voltage level of the data line when driven, which is not a real-world representation of the functionality of the data-line. Violations of driven data-lines should be detectable in signal quality tests.

Measurement of EL_9: Test_J, Test_K and SE0 is still a requirement for certification. EL_9 is defined in the USB 2.0 Test Specification and measures the data line voltage when not driven. EL_9 allows measurement of SE0 and common mode voltage violations. Consequently, all USB controllers must continue to support the Test_J, Test_K and SE0_NAK commands as described in Section 7.1.20 "Test Mode Support."

[↑ Top](#)

Rise and fall time measurements for USB compliance

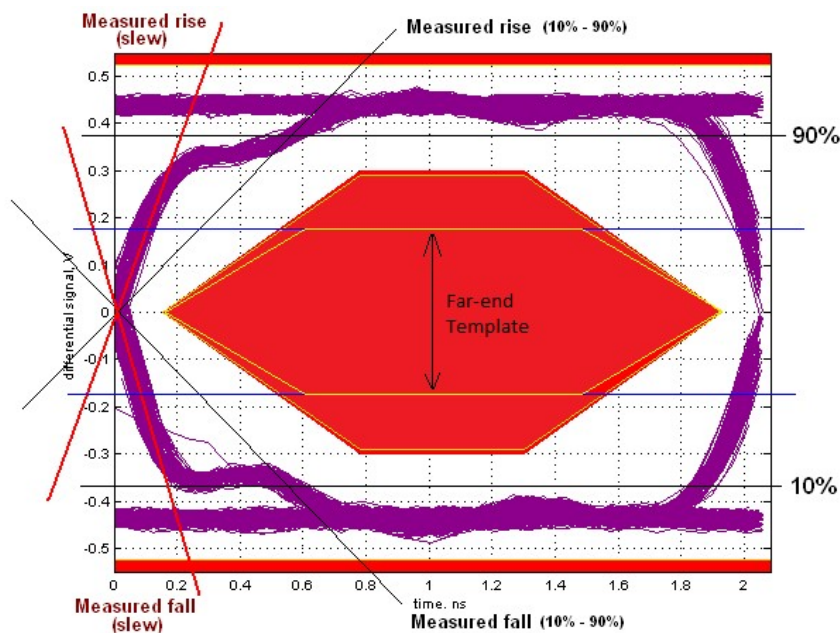
Mandate: Required

Effective Date: August, 2007

There has always been a problem accurately measuring rise and fall times, especially on high speed devices. The measurement of interest is the edge rate, or slew rate, during the state change time. To help improve accuracy of the measurement, the USB-IF is standardizing on one [test fixture for high-speed signal quality](#).

Aside from the fixturing and probes used to take the measurements, major contributors to the inaccuracies in these measurements are the shape of the edge, noise on the signal and the method of calculating the 10% and 90% points as defined in Sections 7.1.2.1 and 7.1.2.2 of the USB 2.0 Specification.

A waveform with slow corners (see sample eye diagram below) will result in a measured rise time that is slower than the actual edge rate would indicate. Also a small change in the position of the 10% and 90% points due to noise on the signal, etc., can cause a relatively large change in the measured rise time.



In order to optimize the repeatability of this measurement a slew rate, or edge rate, will be used. The results are expressed in terms of volts per micro-second (V/us). The conversion from rise time to slew rate uses the specified rise time over 80% of the nominal peak to peak signal amplitudes for each of the signaling rates.

High Speed:

Edge Rate(max-nominal) = $(0.8V * 0.8) / 300e-12 = 2133V/us$ (Volts per nano-second)
 Edge Rate(max-warning) = $(0.8V * 0.8) / 100e-12 = 6400V/us$ (Volts per nano-second)

Full Speed:

Edge Rate(min) = $(3.3V * 0.8) / 20e-9 = 130V/us$
 Edge Rate(max) = $(3.3V * 0.8) / 4e-9 = 660V/us$

Low Speed:

Edge Rate(min) = $(3.3V * 0.8) / 300e-9 = 8.8V/us$
 Edge Rate(max) = $(3.3V * 0.8) / 75e-9 = 35.2V/us$

The actual slew rates are measured within the limits of the inner vertical eye of the far-end template for each test type. This method eliminates the subjective location of the vertical measurement points of the 10/90 method since a fixed voltage range is used.

Note that the calculations use the [relaxed edge rate values](#) of 300ps and 100ps for high-speed signaling. Calculating slew rates in USBET20 provides a common measurement method and should result in more consistent results between test equipment manufacturers, test facilities and USB-IF compliance workshops.

This measurement continues to be informational for FS and LS. However, this measurement is required for HS.

The current version of the USBET20 tool converts the slew rate into an equivalent rise and fall time.



[Top](#)

"Near-end" and "Far-end" explained

Mandate: Informational

Effective Date:

High-speed electrical tests are performed either near-end or far-end depending on the configuration of the product. The terms "near-end" and "far-end" are based on which end of the cable the test fixture is attached in relation to the device being tested. The point where measurements are made dictates what eye template is used. (Please see Section 7.1.2.2 of the USB 2.0 Specification) The measurement is made either "near" the device or "far" from the device being tested.

All HS peripherals with a B-receptacle are tested near-end (at the peripheral's receptacle). HS devices that have a captive cable are tested far-end (at the end of the captive cable). Unlike full-speed electrical tests, which are always performed far-end, the length of the cable used in HS electrical tests is not important.

High-speed electrical tests of downstream ports on hosts and hubs are always performed near-end.

Upstream ports that have a captive cable are always tested far-end. Peripherals with permanently attached standard A-plugs, such as "thumb" or "key" drives, are considered to be captive (like a mouse or keyboard). The captive cable for thumb drives is, simply, very short. Thus, far-end electrical tests are

performed on these products.



[Top](#)

64-bit HSET is Unable to Initialize and Execute Tests.

Mandate: Informational

Effective Date: September, 2010

In the 32-bit HSET utility, the EHCI driver is automatically swapped out with a proprietary EHCI test driver. This feature is not available in the 64-bit version (at this time). So you will need to manually replace the Windows EHCI driver with the test EHCI driver.

1. Within Device Manager, locate the Enhanced Host Controller located under the Universal Serial Bus controllers
2. Right click and Update the Driver Software
3. Select "Browse the Computer for Driver Software"
4. Select "Pick from a List of Device Drivers"
5. Make sure the "Show compatible hardware" is checked
6. Select the "Intel® USB 2.0 Enhanced Host Controller" then choose "Have Disk"
7. Browse to C:\Program Files\USB-IF Test Suite\EHCI HS Electrical Test Toolkit\EHCI\Drivers (or something similar) and locate the hcdriver.sys file. At this point, the "Intel EHCI Compliance Test Tool" should be displayed in the hardware model window. Select it and choose "Next."
8. Windows Security will issue a warning. Install the driver anyway.
9. The Windows EHCI stack should now be replaced by the "Intel EHCI Compliance Test Tool" driver the next time the PC is booted.
10. Reboot the system. The "Intel EHCI Compliance Test Tool" driver is not signed and Windows will not automatically load an unsigned driver. **Prior to Windows loading, press F8 and disable driver signature enforcement** in order to load the unsigned driver.

To restore the original Windows EHCI driver, uninstall the Intel Test Stack driver and rescan for hardware changes.



[Top](#)

Hi-Speed Signal Quality Test Fixture

Mandate: Required

Effective Date: August, 2013

The USB 2.0 Specification, which was written in the late 1990's, limits the transition time of hi-speed signals to 500ps, minimum. At that time, 500ps edge rates were considered very fast. However, process improvements and shrinkage in silicon design can easily cause hi-speed signals to transition significantly faster than the specification limit. Consequently, the USB-IF has [relaxed the rise and fall time requirements](#) for certification, but measuring these fast transitions are proving very problematic. Fast transitions act like a TDR pulse and signal quality test fixtures available today are not designed for these fast transitions. Also, the various differential probe designs of the scope manufactures react very differently to fast edges giving very different test results.

In order to obtain more consistent test measurements, the USB-IF is standardizing on a single test fixture for measuring hi-speed signal quality. This test fixture uses SMA connectors that eliminate the need to use probes. Studies have shown that eliminating the use of probes and using one test fixture design significantly improves the consistency and accuracy of hi-speed USB signal quality analysis.

The "USB 2.0 Hi-Speed Signal Quality Test Fixtures" are available from the USB-IF [eStore](#). USB cables are provided with the fixture but the SMA cables are not. The USB-IF does not sell or provide SMA cables and it is the vendor's responsibility to obtain them. One recommendation for SMA cables includes the following: Agilent 15443A Matched Cable Pair (or equivalent).

By August 2013, this test fixture will be used for all certification tests of hi-speed capable devices at all compliance workshops and by all authorized test laboratories. Device vendors may continue to use any test solution available that meets their needs, but USB-IF certification tests will be conducted using the new test fixture.



[Top](#)

New Rise and Fall Time Parameters for High-speed Signaling

Mandate: Required

Effective Date: April, 2013

The USB 2.0 Specification defines minimum rise and fall time of high-speed signal transitions as 500ps. For the purposes of USB certification, the USB-IF relaxes the rise and fall time to 100ps minimum. This will mean that developers will no longer have to implement edge rate control gimmickry to meet the rise and fall time requirements of the USB 2.0 Specification in order to obtain USB certification.

A pass value will be assigned to transitions that are 300ps or longer (2133V/us slew rate). For edge rates faster than 300ps, a warning will be issued stating that electromagnetic interference and crosstalk on the data-lines must be considered and mitigated in the product's design. The absolute minimum rise and fall time allowed will be 100ps (6400V/us slew rate). No waivers shall be granted for edge rates faster than 100ps.

[Top](#)

Maximum allowed voltage on non-driven lines of Test_J and Test_K are relaxed

Mandate: Required**Effective Date:** April, 2014

The Hi-Speed Electrical Test Specification, EL_9, measures the voltage on data lines that are not driven during test modes Test_J, Test_K and SE0_NAK. For the purposes of certification, when either D+ or D- is not being driven, the output voltage must be $0V \pm 20mV$ when terminated with precision 45 Ω resistors to ground.

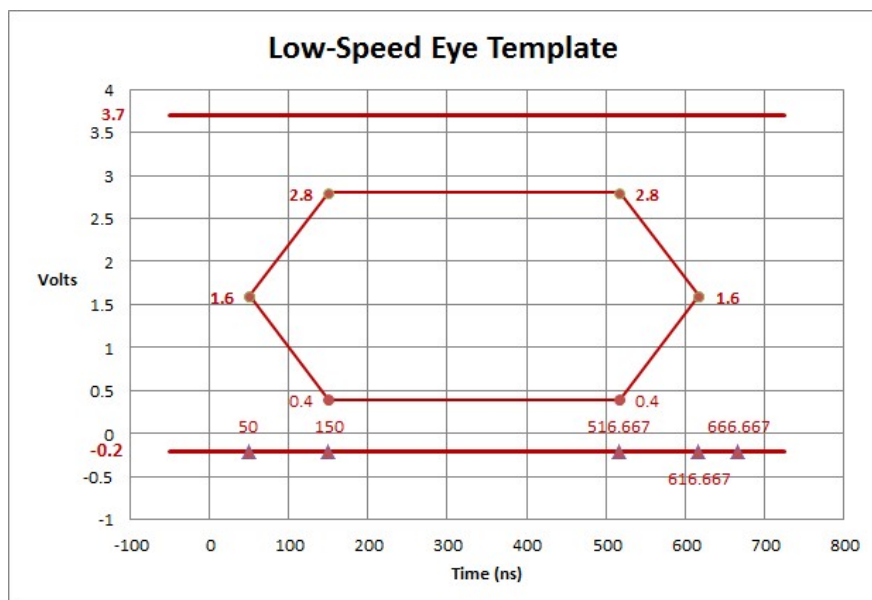
[Top](#)

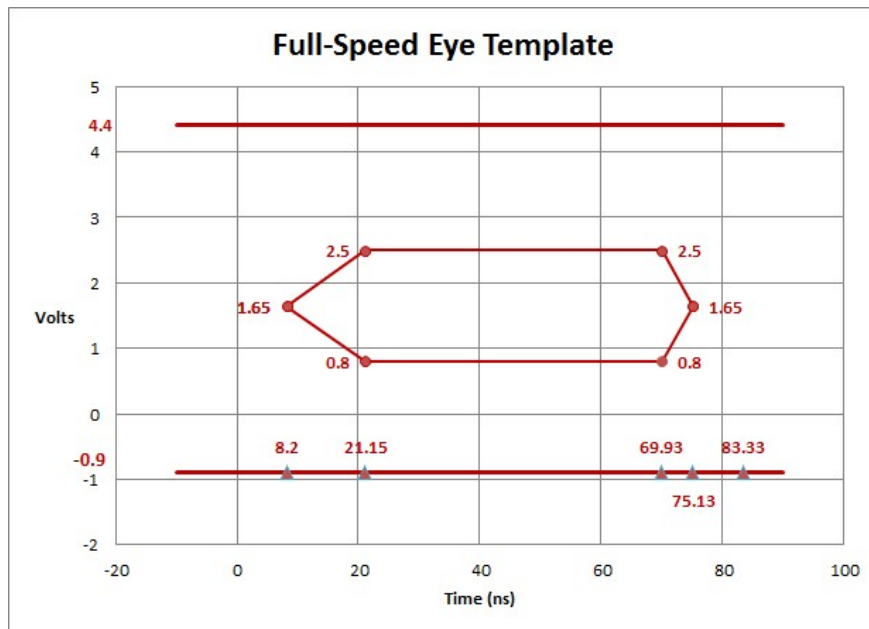
Low-speed and Full-speed Eye Template Definition

Mandate: Informational**Effective Date:**

Section 7.1.2.2 of the USB 2.0 Specification defines the eye templates for hi-speed signaling. The USB 2.0 Specification does not define the eye template for low-speed and full-speed signaling. However, the USB-IF has implemented low-speed and full-speed eye templates for signal quality analysis since the publication of the USB 2.0 Specification in year 2000.

The eye template parameters are derived from the USB 2.0 Specification. Some parameters have been slightly modified over the years as the USB-IF compliance committee made waiver boundaries permanent.





[Top](#)

ECN: Increase to the Maximum Voltage on VBus

Mandate: Required

Effective Date: August, 2014

The USB-IF has published a change to the USB 2.0 Specification that increases the maximum voltage level allowed on VBus to 5.5V. The ECN is included with the download of the USB 2.0 Specification.

[Top](#)

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